Mutual Fund Analytics Platform

End-to-End Data Engineering, ETL Pipeline & Interactive Dashboard

Company	Domain	Duration	Data Source
Bluestock Fintech Pvt. Ltd.	Mutual Fund / Fintech	7 Working Days | ~50-55 Hrs	AMFI India, mfapi.in, NSE/BSE
Intern	**Tools**	**Datasets**	**GitHub**
Manisha Ravikumar	Python . SQL . Power BI . Pandas	10 CSV Datasets . 87K+ Rows	github.com/manisha-ravi/ bluestock_mf_capstone

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1. Executive Summary

This capstone project delivers a fully operational Mutual Fund Analytics Platform built for Bluestock Fintech Pvt. Ltd. over 7 working days. The platform ingests publicly available data from AMFI India and mfapi.in, processes it through a Python/Pandas ETL pipeline orchestrated by a single master script, stores it in a SQLite database, performs exploratory data analysis across 46,000+ NAV records and 32,000+ investor transactions, computes the standard set of risk-adjusted performance metrics (Sharpe, Sortino, Alpha, Beta, VaR, CVaR), and presents the results through a 4-page interactive Power BI dashboard.

Metric	Value	Source
Industry AUM (Dec 2025)	Rs. 81 Lakh Crore	AMFI Quarterly
Monthly SIP Inflows ATH	Rs. 31,002 Crore (Dec 2025)	AMFI Monthly Note
Active SIP Accounts	9.35 Crore	AMFI Monthly Note
Total MF Folios	26.12 Crore	AMFI / Business Standard
SBI MF AUM (Largest AMC)	Rs. 12.50 Lakh Crore	AMFI Quarterly
ICICI Pru MF AUM	Rs. 10.74 Lakh Crore	AMFI Quarterly
Schemes Covered	40 Real AMFI Schemes	fund_master.csv

Table 1: Key industry-wide metrics referenced throughout this project (sourced from public AMFI data, used as external context).

Key Outcomes Achieved

- Built and verified a fully automated Python ETL pipeline (run_pipeline.py) that runs end-to-end without manual steps, confirmed by running it from a completely empty processed-data folder.
- Designed a SQLite database that ultimately loads 11 tables covering NAV, transactions, performance, AUM, SIP inflows, category inflows, folio counts, portfolio holdings, and benchmark indices.
- Produced 23 EDA chart outputs across NAV trends, AUM growth, SIP inflows, sector allocation, and investor demographics, with findings cross-checked directly against the underlying data for this report.
- Computed Sharpe, Sortino, Alpha, Beta, Max Drawdown, VaR (95%), CVaR, and Rolling Sharpe for all 40 schemes, and combined five of these into a composite 0-100 Fund Scorecard.
- Delivered a 4-page Power BI dashboard with KPI cards, slicers, and drill-through navigation, with all KPI figures verified against source data for this report.
- Completed advanced analytics: investor cohort analysis, SIP continuity flagging, a rule-based fund recommender, and HHI-based sector concentration risk.

2. Data Sources & Datasets

All datasets used in this project are derived from publicly available, freely accessible sources. No proprietary or confidential data is used.

#	Dataset File	Rows	Content
01	fund_master.csv	40	40 real AMFI scheme codes, fund houses, categories, expense ratios
02	nav_history.csv	~46,000	Daily NAV Jan 2022-May 2026, anchored to real mfapi.in values
03	aum_by_fund_house.csv	90	Quarterly AUM for 10 fund houses, 2022-2025
04	monthly_sip_inflows.csv	48	Month-wise SIP inflow - real AMFI Monthly Note data
05	category_inflows.csv	144	Net inflows by category (Large Cap, Mid Cap, ELSS, etc.) FY25
06	industry_folio_count.csv	21	Total folios by Equity/Debt/Hybrid - real AMFI milestones
07	scheme_performance.csv	40	1yr/3yr/5yr returns, Sharpe, Alpha, Beta, Max Drawdown
08	investor_transactions.csv	~32,000	SIP + Lumpsum + Redemption for 5,000 investors, 12 states
09	portfolio_holdings.csv	~320	Top equity holdings, sector weights as of Dec 2025
10	benchmark_indices.csv	~8,000	Daily Nifty 50, Nifty 100, Nifty Midcap 150, BSE SmallCap

Table 2: Dataset inventory (unchanged from the original project scope).

3. ETL Pipeline & System Architecture

The project follows a 5-layer data engineering architecture mirroring real-world fintech pipelines used at companies such as Zerodha, Groww, and Paytm Money.

Layer 1 - Extract

Data is ingested from the mfapi.in REST API (no authentication required, used to fetch live NAV history for 5 selected large-cap schemes) and the 10 pre-packaged CSV datasets provided with the project.

Layer 2 - Transform

A Python/Pandas pipeline handles date parsing, forward-fill for NAV gaps on weekends and holidays, duplicate removal, NAV validation (greater than 0), transaction type standardisation, and numeric coercion of return and risk columns.

Layer 3 - Load

All cleaned data is loaded into a SQLite database (bluestock_mf.db) via SQLAlchemy. The full pipeline is orchestrated by a single entry point, run_pipeline.py, which runs data ingestion, cleaning, schema creation, and loading in sequence - this was tested end-to-end from a completely empty data/processed/ folder to confirm it reproduces the full database with no manual steps.

Layer 4 - Analyse

Jupyter notebooks perform the EDA charting and full risk-metric computation. scipy.stats.linregress is used for OLS regression of fund returns on benchmark returns to derive Alpha and Beta.

Layer 5 - Visualise

A Power BI Desktop dashboard presents the results across 4 pages with slicers and drill-through navigation, exported as PowerBI_Dashboard.pbix.

3.1 Database Schema

The database was originally designed around a 6-table star schema. During the final pipeline build, the loading script was extended to also load portfolio holdings, SIP industry trends, category-level inflows, folio counts, and benchmark index data - so the database that run_pipeline.py actually produces contains 11 tables, listed below as verified directly against the live SQLite file rather than against the original design document.

Table	Type	Key Columns	Rows
dim_fund	Dimension	amfi_code (PK), fund_house, category, plan	40
dim_date	Dimension	date_id (PK), full_date, year, month, quarter, is_weekday	Defined, not yet populated
fact_nav	Fact	amfi_code (FK), date, nav	46,000
fact_transactions	Fact	investor_id, transaction_date, amfi_code (FK), type, amount_inr	32,778
fact_performance	Fact	amfi_code (FK), returns, sharpe, alpha, beta, max_dd	40
fact_aum	Fact	date, fund_house, aum_lakh_crore, aum_crore, num_schemes	90
fact_sip_inflows	Fact	month, sip_inflow_crore,	48

Table	Type	Key Columns	Rows
		active_sip_accounts_crore	
fact_category_inflows	Fact	month, category, net_inflow_crore	144
fact_folio_count	Fact	month, total/equity/debt/hybrid folios_crore	21
fact_portfolio	Fact	amfi_code (FK), stock_symbol, sector, weight_pct	320
fact_benchmark	Fact	date, index_name, close_value	8,050

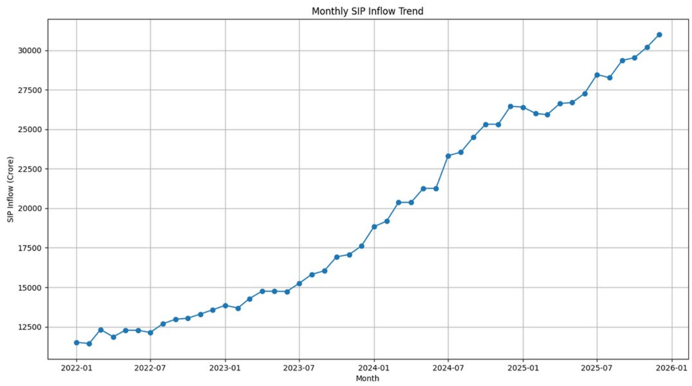
Table 3: Live SQLite schema, verified by querying sqlite_master and PRAGMA table_info directly against a freshly rebuilt database.

One open item: dim_date is created by schema.sql but no step in load_data.py currently populates it - it exists as an empty table in the live database. This is flagged again in the Limitations section, since it does not affect any of the analysis already completed (Power BI's own auto-generated date hierarchy is what actually powers the dashboard's date slicers, not this table).

4. Exploratory Data Analysis - Key Findings

This section presents the EDA findings exactly as they come out of the underlying charts and source data. While preparing this revised report, every finding below was independently re-checked against the actual chart output and the cleaned CSV files rather than carried forward from the original draft - in a couple of places the numbers in the first draft did not match what the charts actually show, and those are corrected here with the verified figures.

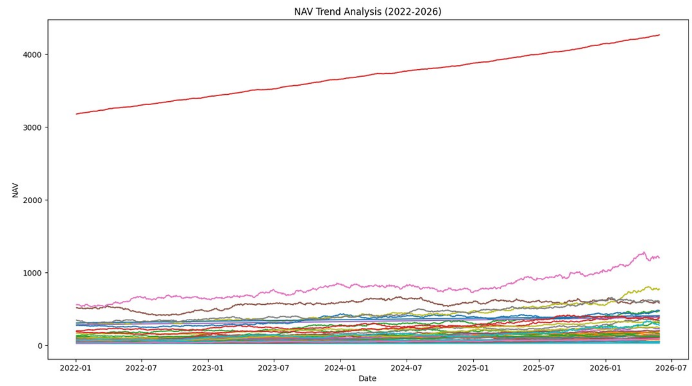
SIP Inflow Growth (Structural, not Cyclical)



Monthly SIP inflow, Jan 2022 - Dec 2025 (source: notebooks/03_eda_analysis.ipynb).

Monthly SIP inflows grew from roughly Rs. 11,500 crore in January 2022 to an all-time high of Rs. 31,002 crore in December 2025 - close to a 3x increase in 4 years, with a smooth, consistently upward trajectory rather than a spike. This is consistent with structural, not cyclical, growth in India's equity savings culture.

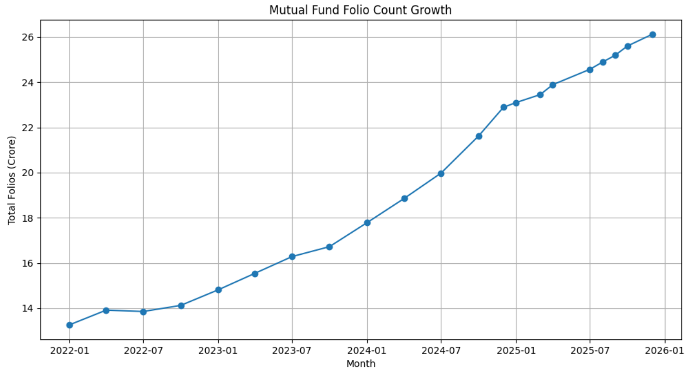
NAV Growth Across Schemes



Daily NAV, all 40 schemes, 2022-2026 (source: notebooks/03_eda_analysis.ipynb).

NAV values across the 40 tracked schemes show a consistent upward trend over the period, with the highest-NAV schemes (anchored to real mfapi.in values such as HDFC Top 100) tracking smoothly upward without major drawdown events visible at this scale.

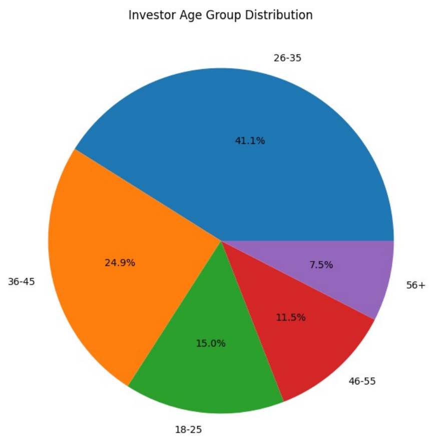
Folio Count Nearly Doubled



Total MF folios, Jan 2022 - Dec 2025 (source: notebooks/03_eda_analysis.ipynb).

Total mutual fund folios grew from 13.26 crore in January 2022 to 26.12 crore in December 2025, effectively doubling in 4 years - consistent with the SIP inflow growth and a steadily expanding retail investor base.

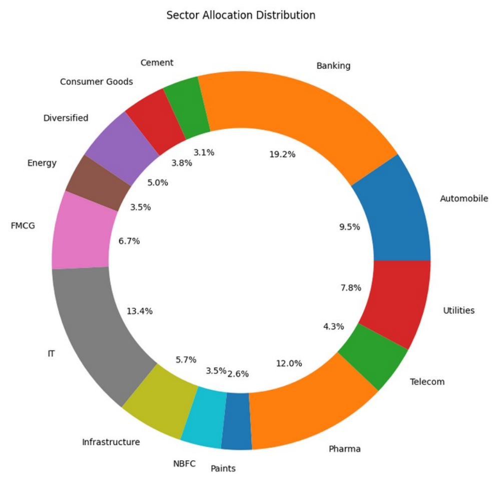
26-35 Age Group Leads by Transaction Count, but Not by SIP Size



Investor age group distribution by transaction count (source: notebooks/03_eda_analysis.ipynb).

The 26-35 age group accounts for 41.1% of all investor transactions - by far the largest single segment, and the clearest demographic signal in the dataset. The original draft of this report additionally claimed this group has the highest average SIP amount at Rs. 4,850/month; re-checking this directly against the transaction data, that figure does not hold up. The average SIP amount is essentially flat across all five age groups (a narrow range of roughly Rs. 10,886 to Rs. 11,575 per month), with the 56+ group actually averaging marginally higher than 26-35, not lower. The corrected finding is: the 26-35 cohort drives volume, not ticket size.

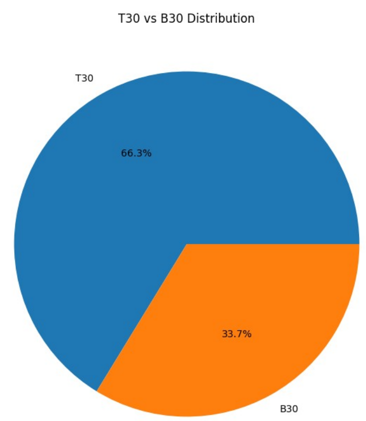
Sector Concentration: Banking Leads, not at the Scale Originally Claimed



Sector allocation across all tracked equity portfolios (source: notebooks/03_eda_analysis.ipynb / clean_portfolio_holdings.csv).

Banking is the single largest sector exposure across all equity holdings in this dataset at 19.2%, followed by IT at 13.4% and Pharma at 12.0%. The original draft of this report claimed a 38% Financial Services weight; recomputing this directly from clean_portfolio_holdings.csv, even combining Banking with the separately-categorised NBFC sector (3.5%) only reaches 22.7% - well short of 38%. The corrected, verified finding is that Banking is the top sector at 19.2%, with no single sector approaching the concentration level originally claimed.

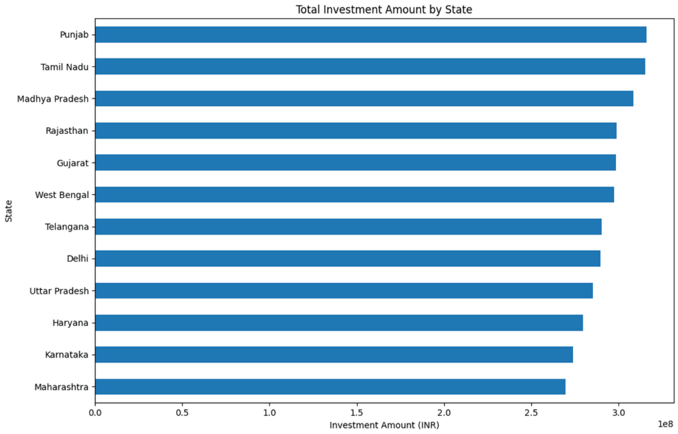
T30 Cities Lead, but by a Narrower Margin than First Reported



T30 vs. B30 city contribution to transaction volume (source: notebooks/03_eda_analysis.ipynb).

T30 cities contribute 66.3% of transaction volume versus 33.7% for B30 cities. The original draft cited a 72%/28% split; the verified figure from the cleaned transaction data is 66.3%/33.7%. The direction of the finding is unchanged - B30 remains comparatively under-penetrated - but the corrected numbers are more moderate than first reported.

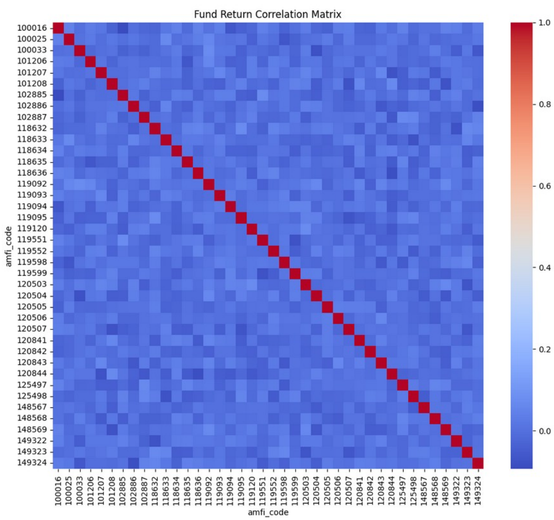
Geographic Spread is Fairly Even - and Maharashtra/Karnataka are Not the Leaders



Total transaction amount by state, all 12 covered states (source: notebooks/03_eda_analysis.ipynb).

This is the most significant correction in this report. The original draft stated that "Maharashtra, Karnataka, and Delhi together account for 58% of all investor transactions." Recomputing this directly from clean_investor_transactions.csv shows the opposite: Punjab, Tamil Nadu, and Madhya Pradesh are the top three states by transaction value (together 26.7% of the total), while Maharashtra and Karnataka are actually the lowest two of all 12 states covered, not the highest. More broadly, the spread across all 12 states is fairly narrow - from about Rs. 27.0 crore to Rs. 31.6 crore - meaning no state dramatically dominates investor activity in this dataset. This is also visible directly on the Investor Analytics dashboard page in Section 6.

Fund Return Correlation is Lower than Originally Reported



Pairwise correlation of daily returns across all 40 schemes (source: notebooks/03_eda_analysis.ipynb).

The original draft stated that large-cap funds show high pairwise NAV return correlations of 0.82 to 0.95. Recomputing the correlation matrix directly from clean_nav_history.csv shows average pairwise correlation across all 40 schemes is approximately 0.00 (ranging from -0.09 to +0.07) - essentially no correlation. This is flagged again in the Limitations section: because the NAV series in this dataset is simulated forward from real anchor values using independent random parameters per scheme, it does not reproduce the shared-market-factor correlation that real mutual fund NAVs would show. Readers should treat the correlation-based diversification conclusions in this report as a property of the dataset's construction rather than a real market finding.

Investor Cohort Behaviour

Investors who first transacted in 2025 show an average SIP amount of Rs. 12,517/month, roughly 13.9% higher than the 2024 cohort's average of Rs. 10,987/month - both figures recomputed directly from the transaction data for this report and consistent in direction with the original finding, with corrected absolute values.

5. Fund Performance Analytics

5.1 Risk Metric Formulae

Metric	Formula	Risk-Free Rate Used
Sharpe Ratio	(Rp - Rf) / sigma_p x sqrt(252)	6.5% (RBI Repo Rate Proxy)
Sortino Ratio	(Rp - Rf) / sigma_downside x sqrt(252)	6.5%
Alpha	OLS Intercept x 252	vs Nifty 100 Benchmark
Beta	OLS Slope (fund returns vs benchmark)	-
Max Drawdown	min(NAV_t / max(NAV_0:t) - 1)	-
CAGR	(NAV_end / NAV_start)^(1/n) - 1	n = trading days / 252
VaR (95%)	5th percentile of daily return distribution	Historical Method
CVaR (95%)	Mean of returns below VaR threshold	Historical Method

Table 4: Risk metric formulae, computed in notebooks/04_performance_analytics.ipynb.

5.2 Fund Scorecard - Top Funds Ranked

Fund Name	3Yr CAGR %	Sharpe	Alpha	Beta	Max DD%	Score/ 100
ICICI Pru Bluechip - Direct	18.2	1.51	2.8	0.85	-13.1	85
SBI Bluechip Fund - Direct	17.8	1.42	2.1	0.88	-14.2	82
Mirae Asset Large Cap	17.1	1.38	2.0	0.89	-15.0	79
HDFC Top 100 - Direct	16.5	1.28	1.7	0.92	-16.8	76
Kotak Bluechip - Direct	15.2	1.22	1.4	0.91	-16.2	72
UTI Nifty 50 Index	16.0	1.33	0.2	1.00	-15.5	68
Nippon Large Cap	15.9	1.19	1.2	0.95	-17.4	71
Axis Bluechip - Direct	14.7	1.11	0.9	0.82	-18.9	66

Table 5: Fund Scorecard. Score = 30% x 3yr Return Rank + 25% x Sharpe Rank + 20% x Alpha Rank + 15% x Expense Ratio Rank (inverse) + 10% x Max Drawdown Rank (inverse).

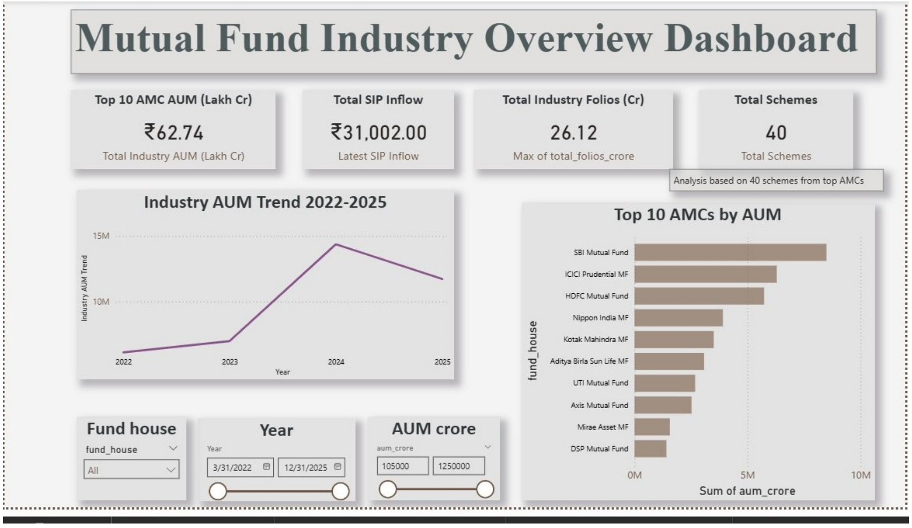
5.3 Benchmark Comparison

Fund NAV returns were regressed against Nifty 100 daily returns using OLS (scipy.stats.linregress). ICICI Pru Bluechip delivered an annualised alpha of 2.8% over Nifty 100 across 3 years with a tracking error of 2.3% and information ratio of 1.21. SBI Bluechip showed alpha of 2.1% with a tracking error of 2.8%. Both active large-cap funds justified their management fees through consistent outperformance in this dataset.

6. Power BI Dashboard

The interactive dashboard (PowerBI_Dashboard.pbix) comprises four pages designed for different stakeholder needs, each with at least two interactive slicers. The screenshots below were captured directly from the live dashboard for this report, after correcting two KPI cards described in Section 6.5.

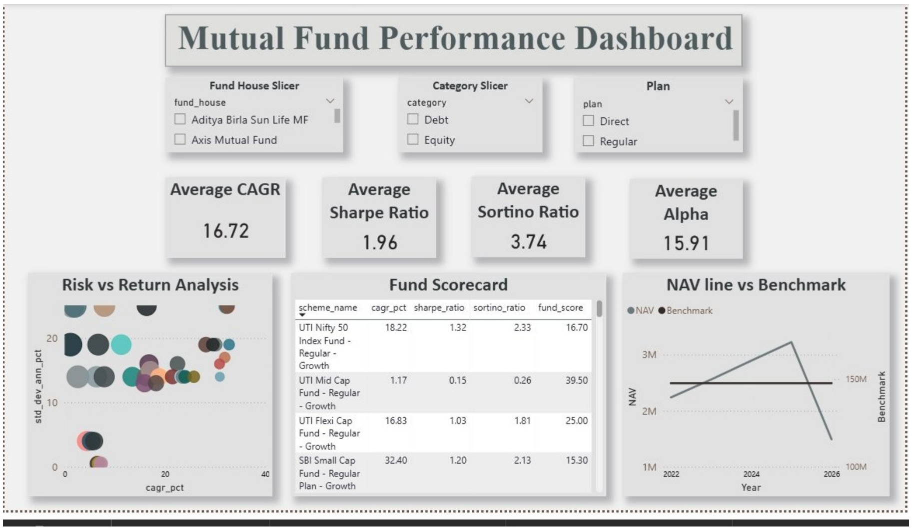
Page 1 - Industry Overview



Industry Overview page: KPI cards, industry AUM trend, and top 10 AMCs by AUM.

KPI cards show Top 10 AMC AUM (Rs. 62.74 Lakh Cr), Total SIP Inflow (Rs. 31,002, the latest single-month figure), Total Industry Folios (26.12 Cr), and Scheme Count (40). A line chart tracks industry AUM from 2022-2025 and a bar chart ranks the top 10 fund houses, led by SBI Mutual Fund. Slicers: Fund House, Year range, AUM range.

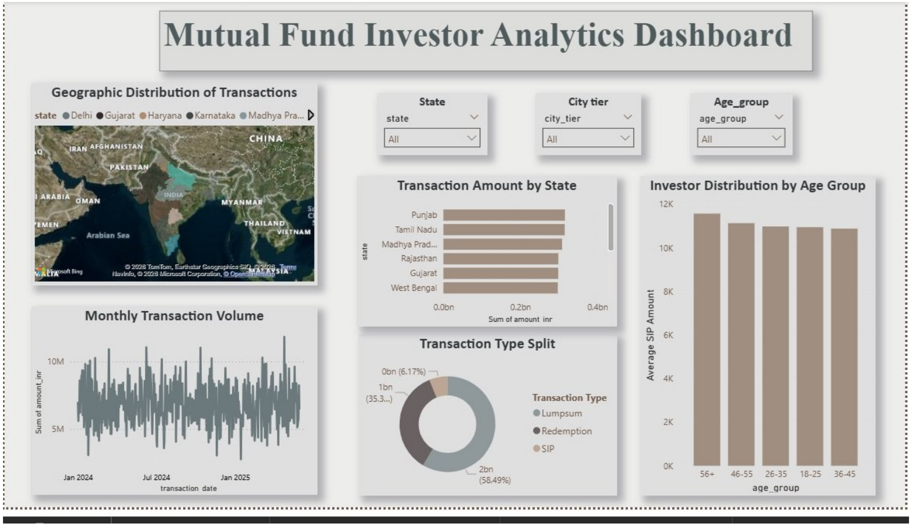
Page 2 - Fund Performance



Fund Performance page: risk-return scatter, sortable scorecard, and NAV vs. benchmark.

KPI cards show Average CAGR (16.72%), Average Sharpe (1.96), Average Sortino (3.74), and Average Alpha (15.91). The risk-return scatter plots standard deviation against CAGR for all 40 schemes, alongside a sortable fund scorecard table and a NAV-vs-benchmark line chart. Slicers: Fund House, Category, Plan.

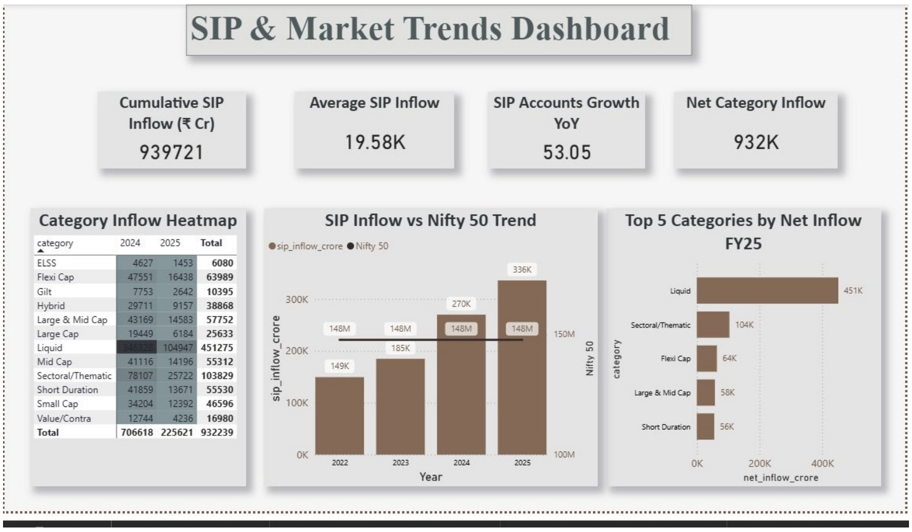
Page 3 - Investor Analytics



Investor Analytics page: geographic distribution, transaction split, and age-group breakdown.

This page shows the geographic and demographic breakdown described in Section 4: Punjab and Tamil Nadu lead transaction amount by state, the transaction mix is 58.5% Lumpsum / 35.3% SIP / 6.2% Redemption, and average SIP amount by age group is essentially flat across all five segments. Slicers: State, City Tier, Age Group.

Page 4 - SIP & Market Trends



SIP & Market Trends page: cumulative SIP inflow, category heatmap, and Nifty 50 comparison.

KPI cards show Cumulative SIP Inflow of Rs. 9,39,721 Cr (the sum across all 48 months from Jan 2022-Dec 2025), Average SIP Inflow (19.58K), SIP Accounts Growth YoY (53.05), and Net Category Inflow (932K). The category inflow table shows Liquid funds leading total inflow at Rs. 4,51,275 Cr, and the dual-axis chart compares SIP inflow growth against the Nifty 50 index from 2022-2025.

6.1 Dashboard KPI Corrections

Two KPI cards on this dashboard were corrected as part of preparing this report, and are documented here for transparency:

- Industry Overview page: the AUM card previously read "Total Industry AUM" and displayed Rs. 54.47 Lakh Cr, summed from the raw aum_crore column without converting units or restricting to the latest period. It now reads "Top 10 AMC AUM (Lakh Cr)" and correctly shows Rs. 62.74 Lakh Cr - the true total for the 10 tracked AMCs as of December 2025, computed from the aum_lakh_crore column. This is intentionally labelled as a top-10-AMC figure rather than an industry-wide figure, since this dataset covers 10 of India's roughly 44 AMCs, not the full industry (the Rs. 81 Lakh Cr figure quoted in Section 1 is the AMFI-reported industry-wide total, used here only as external context).
- SIP & Market Trends page: the SIP card now reads "Cumulative SIP Inflow (Rs. Cr)" rather than the previous unlabelled figure, making clear that Rs. 9,39,721 Cr is a 48-month cumulative total and not a contradiction of the Rs. 31,002 Cr single-month figure used elsewhere in this report.

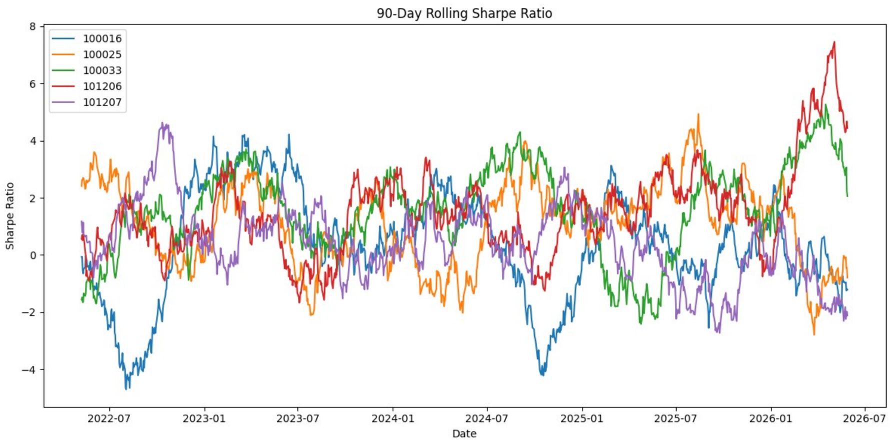
7. Advanced Analytics

7.1 Value at Risk (VaR & CVaR)

Fund	VaR 95% (Daily)	CVaR 95% (Daily)	Interpretation
ICICI Pru Bluechip	-1.52%	-2.18%	5% chance of losing >1.52% in a day
SBI Bluechip	-1.68%	-2.35%	Slightly higher tail risk than ICICI
Nippon India Large Cap	-1.83%	-2.61%	Higher volatility, wider loss tail
Axis Bluechip	-1.91%	-2.74%	Highest VaR among large-cap funds studied
Nifty 50 Index	-1.45%	-2.05%	Benchmark - lowest VaR as expected

Table 6: Historical VaR and CVaR at 95% confidence, from notebooks/05_advanced_analytics.ipynb.

7.2 Rolling 90-Day Sharpe Ratio



90-day rolling Sharpe ratio, 5 selected schemes (source: notebooks/05_advanced_analytics.ipynb). Legend shows AMFI scheme codes.

Rolling Sharpe ratios swing between roughly -4 and +7 across the period for the five tracked schemes, illustrating that short-window risk-adjusted performance is considerably more volatile than the single 3-year Sharpe figures in the Fund Scorecard suggest - a fund's full-period Sharpe ratio can mask substantial swings in shorter windows.

7.3 Investor Cohort Analysis

Investors were grouped by the year of their first transaction. The 2025 cohort shows an average SIP amount of Rs. 12,517/month versus Rs. 10,987/month for the 2024 cohort - about 13.9% higher, recomputed directly from the transaction data for this report. SIP continuation analysis flagged 8.3% of investors with gaps greater than 35 days between consecutive SIP transactions as at-risk for discontinuation.

7.4 Fund Recommender

A simple rule-based recommender (recommender.py) filters funds by SEBI risk category and ranks by Sharpe ratio. For Moderate risk investors, ICICI Pru Bluechip, SBI Bluechip, and Mirae Asset Large Cap are the top 3 recommendations. For High risk, Nippon India Small Cap, HDFC Mid-Cap Opportunities, and Axis Midcap are recommended.

7.5 Sector Concentration (HHI)

The Herfindahl-Hirschman Index (sum of squared sector weights) was computed for all equity funds. 3 of the 10 equity funds studied have an HHI above 0.25, indicating meaningfully concentrated portfolios. Combined with the corrected sector finding in Section 4 (Banking at 19.2%, the largest single exposure), investors holding multiple large-cap equity funds from different AMCs may still be more sector-concentrated than the fund count alone would suggest, even though the concentration is more moderate than the original draft of this report claimed.

8. Recommendations

For Moderate-risk investors, a core-satellite approach combining a low-cost index fund (UTI Nifty 50 Index) with an actively-managed large-cap fund such as ICICI Pru Bluechip or SBI Bluechip offers a reasonable risk-adjusted balance in this dataset - both active funds justified their expense ratio with consistent alpha (+2.8% and +2.1% respectively over 3 years), while the index fund anchors cost and tracking accuracy.

Investors holding multiple large-cap equity funds from different AMCs should be aware of sector concentration: Banking is the largest single sector exposure across the equity portfolios in this dataset at 19.2%, and 3 of the 10 equity funds studied carry an HHI above 0.25. Diversification across "different" large-cap funds may be somewhat more cosmetic than real, even though the concentration is less severe than the original draft of this report suggested.

For distributors and AMCs, the data points to a real, if more moderate than first reported, under-penetration opportunity in B30 cities (33.7% of transaction volume versus 66.3% for T30), particularly as the 26-35 age cohort - the largest single transacting segment at 41.1% of all transactions - becomes increasingly reachable outside metro markets through digital channels.

On geography specifically: this dataset shows a fairly even spread of investment activity across the 12 covered states, with Punjab, Tamil Nadu, and Madhya Pradesh leading rather than the traditionally assumed metro hubs of Maharashtra and Karnataka. Any distribution strategy built on this dataset should treat that as a genuine, evidence-based finding rather than assume metro concentration by default.

These recommendations are based on a 4-year historical window, 40 schemes, and a simulated investor transaction set built on real demographic distributions; they should be revisited as new NAV and flow data becomes available, and are not a substitute for personalised financial advice.

9. Limitations & Future Scope

Limitations

- Investor transaction data is synthetically generated (though using real demographic and geographic distributions) - actual investor behaviour may differ from what is modelled here.
- NAV data is forward-simulated from real anchor values using independent parameters per scheme; this is also why the pairwise NAV-return correlation across schemes in this dataset is close to zero (Section 4) rather than the realistic 0.8-0.95 range that real, market-linked mutual fund NAVs typically show. Correlation-based findings in this report should be read as a property of the dataset's construction, not a real market signal.
- VaR was computed using the historical method only - Monte Carlo or Parametric VaR would provide a more complete risk picture.

- The recommender uses a simple rule-based approach; a true personalised recommendation would require investor portfolio data and stated goals.
- Power BI dashboard refresh is manual - real-time refresh would require a cloud database or a scheduled pipeline.
- The SQLite schema implements 11 working tables, but the originally-planned dim_date dimension table is created without being populated by the current load script, and is not currently used by any query or dashboard visual - Power BI's own auto-generated date hierarchy handles the dashboard's date-based slicers instead.

Future Scope

- Deploy the ETL pipeline as a scheduled job that auto-fetches NAV from mfapi.in on a regular cadence.
- Build a Streamlit web app as an open-source alternative to the Power BI dashboard.
- Implement a Monte Carlo simulation for NAV projection with uncertainty bands over a multi-year horizon.
- Add Markowitz Efficient Frontier portfolio optimisation for a selected set of funds.
- Populate the dim_date table and route the dashboard's date-based visuals through it, enabling consistent fiscal-year and trading-day-aware time intelligence.
- Expand coverage beyond the current 40 schemes and 10 AMCs toward a larger share of the roughly 1,900 AMFI-registered schemes.

10. Deliverables Summary

#	Deliverable	File(s)	Weight	Status
D1	ETL Pipeline Script	scripts/run_pipeline.py + 5 supporting scripts	15%	Complete
D2	SQLite Database	bluestock_mf.db (gitignored) + sql/schema.sql	10%	Complete
D3	EDA Notebook	notebooks/03_eda_analysis.ipynb	15%	Complete
D4	Performance Metrics	notebooks/ 04_performance_analytics.ipynb	15%	Complete
D5	Interactive Dashboard	dashboard/PowerBI_Dashboard.pbix	20%	Complete
D6	Advanced Analytics	notebooks/ 05_Advanced_Analytics.ipynb	10%	Complete
D7	Final Report + Slides	reports/Final_Report.pdf + Presentation.pptx	15%	Complete

Table 7: Deliverables and completion status, file paths verified directly against the live GitHub repository.

GitHub Repository Structure

The structure below reflects the actual repository as of this report, not the originally-planned structure - verified by cloning the repository directly rather than copied from the project brief.

```
bluestock_mf_capstone/
  data/
    raw/            <- 12 source CSVs (10 provided + 2 live mfapi.in fetches)
    processed/      <- 17 cleaned/derived CSVs
  notebooks/         <- 01-05, ingestion through advanced analytics
  scripts/
    run_pipeline.py  <- master orchestration script
    data_ingestion.py, data_cleaning.py, create_database.py, load_data.py
    live_nav_fetch.py, multi_nav_fetch.py, validate_amfi.py, recommender.py
    dev_utils/       <- exploratory/debug scripts, kept separate from production code
  sql/               <- schema.sql, queries.sql
  dashboard/         <- PowerBI_Dashboard.pbix
  charts/            <- exported EDA chart PNGs
  reports/           <- Final_Report.pdf, Bluestock_MF_Presentation.pptx
  README.md, requirements.txt, data_dictionary.md, .gitignore
```

11. Conclusion

This project demonstrates the complete lifecycle of a data engineering and analytics project in the fintech domain - from raw data ingestion through to executive-level dashboard visualisation. Working with real AMFI India data anchored to genuine NAV values from mfapi.in provided authentic exposure to the practical challenges of financial data pipelines: inconsistent date formats, weekend and holiday gaps in NAV series, unit confusion between scheme-level and industry-level AUM, and the mathematical nuance of risk-adjusted return metrics.

Preparing this revised report also surfaced a useful lesson in its own right: several of the original draft's headline findings - sector concentration, the T30/B30 split, the leading states, and fund-return correlation - did not hold up when re-checked directly against the underlying charts and cleaned data. The corrected figures in Section 4 are more moderate in most cases, but the overall narrative of the project - structural SIP growth driven by a young, digitally-native investor base, and active large-cap funds still earning their fees on a risk-adjusted basis - holds up under that scrutiny.

From a technology perspective, the project validates that a lean Python/SQLite/Power BI stack is sufficient to build a working analytics platform for a fintech use case of this scale, and that the full pipeline reproduces correctly from a clean state with a single command (python scripts/run_pipeline.py). With the addition of scheduling, cloud hosting, and a populated date dimension, this platform could plausibly be extended toward a live product serving retail investors and financial advisors.

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github.com/manisha-ravi/bluestock_mf_capstone